NSE INVEST UAT

37 Reporting APIs - Technical Testing, Response Evidence and Backend Handoff

Prepared for backend developer review and Amazon Q code audit | Based on recorded UAT test-session evidence through 14 October 2026

Measure	Recorded result
Reporting POST routes registered	37 / 37
Successful NSE application-level responses	34 / 37
Pending NSE validation	3 / 37
Backend health	nse-api.service active; GET /health -> HTTP 200
Order data available for UAT	0 local stored orders; 0 orders with NSE order IDs

Evidence and limitations

This document is compiled from terminal outputs, code excerpts, and the established test-session ledger provided in the conversation. It is NOT a fresh independent scan of the server or repository. Exact response JSON for each of the 34 accepted APIs was not retained in the supplied evidence; a shared observed response pattern is recorded instead. Do not treat empty successful responses as proof of populated-data correctness.

No credentials, PAN, real client code, tokens, private keys or authentication secrets are included. Sensitive header values must remain redacted during further review.

1. Executive technical status

All 37 reporting routes appear in the live FastAPI /openapi.json output. The service was active and health endpoint returned HTTP 200. 34 report endpoints returned an NSE response_status of S, commonly with report_data_total 0 and error_remark "No record(s) found." Three routes are present and reachable but not validated successfully at NSE application level.

Result class	Meaning for developers
PASS - request accepted	HTTP 200 plus NSE response_status S; usually empty report_data. Authentication, route and supplied parameter combination were accepted in UAT.
PENDING - upstream rejected	HTTP 200 but NSE response_status F. HTTP 200 alone is not a business-level success.
PENDING - no qualifying data	Order-wise report returned F / No record(s) found; cannot distinguish data absence from other business validation without NSE clarification.
NOT TESTED / NOT PROVEN	Populated results, full response mapping, pagination, boundary cases, production behavior and all possible parameter combinations.

2. Complete 37-endpoint testing inventory

All endpoints below use POST. "S / empty" indicates the recorded common successful NSE response, not a verified populated report. Parameter descriptions reflect tested request shapes, not necessarily the complete NSE specification.

#	Local API route	Status	Tested request / constraint	Observed result
1	/api/v1/reports/2fa	PASS - S	DD-MM-YYYY dates	S; generally 0 records
2	/api/v1/reports/allotment-statement	PASS - S	DD-MM-YYYY dates	S; generally 0 records
3	/api/v1/reports/aof-image-upload	PASS - S	client_code	S; generally 0 records
4	/api/v1/reports/client-authorization	PASS - S	DD-MM-YYYY dates	S; generally 0 records
5	/api/v1/reports/client-detail	PASS - S	ISO dates + client_code	S; generally 0 records
6	/api/v1/reports/client-kyc	PASS - S	client_code	S; generally 0 records
7	/api/v1/reports/client-master	PASS - S	client_code	S; generally 0 records
8	/api/v1/reports/elog-upload	PASS - S	client_code	S; generally 0 records
9	/api/v1/reports/fatca	PASS - S	configured TEST_PAN	S; generally 0 records
10	/api/v1/reports/mandate-status	PASS - S	ISO dates; no client_code	S; generally 0 records
11	/api/v1/reports/master-download	PASS - S	file_type=NAV; text file download	S; downloadable text (NAV)
12	/api/v1/reports/member-fund-allocation/age-wise	PASS - S	date=14-10-2026; settlement_type=ALL; amc_code=ALL	S; generally 0 records
13	/api/v1/reports/member-fund-allocation/order-wise	PENDING - F	DD-MM-YYYY dates; NSE status F; no records	F; No record(s) found
14	/api/v1/reports/order-lifecycle	PASS - S	DD-MM-YYYY dates + client_code	S; generally 0 records
15	/api/v1/reports/order-status	PASS - S	ISO dates; required filters ALL	S; generally 0 records

#	Local API route	Status	Tested request / constraint	Observed result
16	/api/v1/reports/provisional-orders	PASS - S	ISO dates; required filters ALL	S; generally 0 records
17	/api/v1/reports/redemption-payout	PENDING - F	report_type rejected by NSE	F; report_type is not valid
18	/api/v1/reports/redemption-payout-non-demat	PENDING - F	report_type rejected by NSE	F; report_type is not valid
19	/api/v1/reports/redemption-statement	PASS - S	DD-MM-YYYY dates	S; generally 0 records
20	/api/v1/reports/securities-delivery-shortage	PASS - S	DD-MM-YYYY dates + client_code	S; generally 0 records
21	/api/v1/reports/sip-cancellation	PASS - S	client_code	S; generally 0 records
22	/api/v1/reports/sip-installment-due	PASS - S	client_code	S; generally 0 records
23	/api/v1/reports/sip-registration	PASS - S	client_code	S; generally 0 records
24	/api/v1/reports/sip-topup	PASS - S	client_code	S; generally 0 records
25	/api/v1/reports/stepup-registration	PASS - S	client_code	S; generally 0 records
26	/api/v1/reports/stp-cancellation	PASS - S	client_code	S; generally 0 records
27	/api/v1/reports/stp-installment-due	PASS - S	client_code	S; generally 0 records
28	/api/v1/reports/stp-registration	PASS - S	client_code	S; generally 0 records
29	/api/v1/reports/swp-cancellation	PASS - S	client_code	S; generally 0 records
30	/api/v1/reports/swp-installment-due	PASS - S	client_code	S; generally 0 records
31	/api/v1/reports/swp-registration	PASS - S	client_code	S; generally 0 records
32	/api/v1/reports/transaction-detail	PASS - S	DD-MM-YYYY dates + client_code	S; generally 0 records
33	/api/v1/reports/xsip-cancellation	PASS - S	client_code	S; generally 0 records
34	/api/v1/reports/xsip-installment-due	PASS - S	client_code	S; generally 0 records
35	/api/v1/reports/xsip-registration	PASS - S	client_code	S; generally 0 records
36	/api/v1/reports/xsip-topup	PASS - S	client_code	S; generally 0 records
37	/api/v1/sip-amc-pause/report	PASS - S	configured client_code ONLY; dates rejected	S; generally 0 records

3. Success and failure response evidence

3.1 Typical accepted report response

```
{ "response_status": "S", "report_data_total": 0, "error_remark": "No record(s) found." }
```

The exact shape and numeric/string type of report_data_total may vary by endpoint; the fragment above summarizes observed output fields and is not a guaranteed full payload. For provisional-orders, the final retest explicitly returned HTTP 200, response_status S and error_remark "No record(s) found."

3.2 Redemption Payout - pending

Local: POST /api/v1/reports/redemption-payout | NSE upstream: /nsemfdesk/api/v2/reports/REDEMPTION_PAYOUT

Observed: HTTP 200; NSE response_status F; error_remark "report_type is not valid." Tried report_type values D, ALL, A, and empty string with slash-separated dates in previous tests. No accepted value was established.

3.3 Redemption Payout Non-Demat - pending

Local: POST /api/v1/reports/redemption-payout-non-demat | NSE upstream: /nsemfdesk/api/v2/reports/REDEMPTION_PAYOUT_NON_DEMAT

Observed: NSE response_status F; "report_type is not valid" for report_type D. Do not infer that all the other values tried on the standard endpoint were also tested on Non-Demat.

3.4 Member Fund Allocation Order-Wise - pending

Local: POST /api/v1/reports/member-fund-allocation/order-wise

```
{ "from_date": "01-09-2026", "to_date": "30-09-2026" }
```

Observed: HTTP 200; response_status F; report_data_total 0; error_remark "No record(s) found" (without a final period). October date ranges also returned F. ISO dates produced HTTP 400 and "Invalid date format. Please use dd-MM-yyyy format." The database contained zero stored order_transactions and zero NSE order IDs, so there was no local order to use for data-based verification.

3.5 Master Download - special successful behavior

Local: POST /api/v1/reports/master-download; tested file_type NAV. NSE returned a pipe-delimited text download (~3,605,217 bytes), not a conventional JSON report. The backend router was changed to handle plain-text file responses as downloadable .txt while retaining JSON handling. A previously attempted SCHEME_MASTER file_type was invalid. Confirm exact response headers, content disposition, MIME type and streaming behavior from the current source; they were not captured in the provided logs.

3.6 Other request validation findings

- Member Fund Allocation Age-Wise accepted date 14-10-2026 with settlement_type ALL and amc_code ALL; omitting either tested required filter led to NSE rejection.

- SIP AMC Pause report accepted only configured client_code in the tested request; adding DD-MM-YYYY or ISO date fields was rejected.

- Order Status and Provisional Orders accepted ISO date fields with relevant filters set to ALL.

- Different report families require different date formats. Do not globally normalize all reports to one date format.

4. Authentication, request headers and code changes

4.1 Confirmed from source excerpt: Redemption Payout router

routers/redemption_payout.py imports get_auth_headers() from auth.nse_auth and get_nse_client() from http_client. It sends an authenticated HTTP POST with json=payload.model_dump(exclude_none=True). This is confirmed by the pasted code; the exact header names and values returned by get_auth_headers() were NOT included in the supplied excerpt.

Representative excerpt (not a complete implementation):

```
response = await client.post(url, headers=get_auth_headers(), json=payload.model_dump(exclude_none=True))
```

The router logs the NSE HTTP status and first 500 characters of the upstream response body; handles HTTP 403 as authentication failure; non-200 as HTTPException; non-JSON as 502; and request timeout as 504. Review whether logging upstream bodies risks exposing sensitive fields.

4.2 Confirmed local test headers

The local Python smoke tests sent Content-Type: application/json to http://127.0.0.1:8000. The upstream NSE authorization headers are generated separately by get_auth_headers(). The recorded UAT session indicates NSE Basic authentication worked, but the exact authorization implementation and any header changes cannot be reconstructed from the supplied evidence.

4.3 Confirmed functional change: Master Download

routers/misc_reports.py was updated to accept NSE text downloads (pipe-delimited NAV data) in addition to JSON responses. The service was restarted and the download path tested. A backup routers/misc_reports.py.bak was noted. Exact diff, response header handling, content-disposition filename and byte-stream safeguards need repository inspection.

4.4 What is NOT confirmed

- No before/after diff for auth.nse_auth or http_client was supplied. Do not claim that Authorization, Accept, User-Agent or custom NSE headers were changed.

- No exhaustive per-endpoint headers, NSE correlation IDs, exact raw bodies or timestamped HTTP traces were supplied.

- No verification that Basic auth, timeout policy and HTTP client behavior are consistent across all 37 routers.

5. Backend repository / infrastructure handoff

Item	Recorded detail
Project directory	/root/NSE_NEW
Python environment	/root/NSE_NEW/venv
Systemd service	nse-api.service
Local API base	http://127.0.0.1:8000
Health endpoint	GET /health -> 200
OpenAPI endpoint	GET /openapi.json -> 37 reporting POST routes
NSE UAT host	https://nseinvestuat.nseindia.com
Relevant modules	routers/redemption_payout.py; schemas/redemption_payout.py; routers/misc_reports.py; auth/nse_auth.py; http_client.py; config.py
DB model	models/order.py: OrderTransaction; table order_transactions
Issue note	/root/NSE_NEW/NSE_PENDING_API_ISSUES.txt
Backups recorded	/root/NSE_BACKUPS/nse_backup_20261014_121043.tar.gz; routers/misc_reports.py.bak

6. Developer review checklist

- Compare each of the 37 local OpenAPI request schemas with official NSE UAT endpoint contracts and supported enum values.

- Inspect auth/nse_auth.py and http_client.py; list exact header names, authentication method, timeout/retry behavior, connection reuse and TLS settings. Redact credentials in all output.

- Diff routers/misc_reports.py against its backup; verify NAV text-download handling, content-type, content-disposition, memory limits and JSON error passthrough.

- Inspect redemption_payout.py and schemas/redemption_payout.py; obtain documented report_type enums for both upstream endpoints before changing validation.

- Ask NSE whether Order-Wise response_status F + "No record(s) found" is expected for empty data; request a known valid UAT order and matching date range.

- Validate date format separately per endpoint; test required vs optional parameters and omission vs ALL filters.

- Capture sanitized, reproducible sample requests and response envelopes for all 37 APIs; add tests for HTTP success with NSE response_status F.

- For the 34 accepted routes, validate populated records, schema parsing, large results, pagination, and failures once UAT data exists.

- Protect PAN, client codes, Basic auth, environment variables, private keys and sensitive response logs; rotate any previously exposed credentials.

- Avoid running purchase/order-creation tests to manufacture data without explicit authorization.

7. Copy-ready Amazon Q audit prompt

Audit the NSE Invest UAT FastAPI backend at /root/NSE_NEW. There are 37 registered reporting POST endpoints. Historical testing recorded 34 NSE response_status=S responses (mostly empty) and 3 pending: redemption-payout, redemption-payout-non-demat (invalid report_type), and member-fund-allocation/order-wise (response_status=F, No record(s) found). Review all 37 route handlers, schemas, NSE upstream URLs, HTTP methods, auth headers generated by auth/nse_auth.py, http_client settings, request field mappings, date formats, optional/required filters, error mapping and logging. Produce a table with exact file:line references, documented/observed request payloads, sanitized header names (NEVER values), and response parsing. Compare routers/misc_reports.py to routers/misc_reports.py.bak to verify the NAV text download change. Do not invent valid NSE report_type values: flag official-spec gaps. Distinguish HTTP 200 from NSE application-level S/F. Suggest safe code fixes and read-only tests, but do not place, cancel, or modify financial transactions. Do not print secrets, PAN or real client codes. Return findings ranked by severity with reproducible test commands and a separate list of questions for NSE support.

8. NSE technical support questions

- What exact report_type values and date format are accepted by REDEMPTION_PAYOUT and REDEMPTION_PAYOUT_NON_DEMAT? Please provide successful UAT JSON examples for each.

- For MEMBER_FUND_ALLOCATION order-wise, does response_status F with No record(s) found represent a valid empty result or a failure?

- Can NSE provide one non-sensitive UAT sample order, associated client/date range, and expected report output?

- Are there report-specific mandatory headers, version headers, required filters or response formats not reflected in the current implementation?

9. Sign-off criteria

Current sign-off is limited to route registration, backend health, and observed acceptance of 34 UAT reporting requests. Do not claim 37/37 fully validated or production ready. Final sign-off requires NSE documentation for the two payout endpoints, a clarified Order-Wise outcome, and representative populated-data tests.